

VELMÈRE PRO AUDIT REPORT

Arbitrum Delayed Inbox (0x4dbd4fc535bd2916850904481a1be141421bd113)

VELMERE SECURITY AUDIT REPORT: ARBITRUM DELAYED INBOX

Network: Ethereum Mainnet (Chain ID: 1)

Report ID: undefined | Tier: PRO | Application Surface: CANONICAL

Created At: 2026-09-09T19:29:42.070Z

VERDICT SUMMARY:

ELEVATED RISK (UNVERIFIED CODE) (72/100)

Release Decision: BLOCKED | Verification Status: INSUFFICIENT_EVIDENCE

STOP-SELL DIRECTIVE: INSUFFICIENT_EVIDENCE: Coverage below minimum threshold for institutional sale

Confidence Score:	60/100	Evidence Coverage:	20%
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Merkle Root: sha256:7dbcebe3cb8e271dbd29d77c923d5078827d987090548e800485149dab5eed07

Contract lacks verified on-chain runtime bytecode. In accordance with Velmère evidence standards, missing bytecode results in an elevated risk assessment (72/100).

AUDIT OVERVIEW & CONTRACT CONTEXT

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Summary of execution scope and target objectives

Project / Contract: Arbitrum Delayed Inbox

Contract Address: 0x4dbd4fc535bd2916850904481a1be141421bd113

Network / Chain ID: Ethereum Mainnet (ID: 1)

Token / Symbol: ARB-INBOX

Proxy Pattern: unverifiable

Compiler Version: Unverified Source

SOURCE CODE VERIFICATION & BYTECODE DECOMPIlation

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

State hash match against blockchain node and dispatcher analysis

EVM machine bytecode unavailable. Disassembly, selector analysis, and privileged opcode scans cannot proceed.

BASELINE SECURITY FINDINGS

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Identified vulnerabilities in baseline scan

Findings & Vulnerability Analysis:

HIGH

VLM-EVM-NO-CODE: Unverified On-Chain Code / Missing Bytecode

Category: Missing Bytecode Truth | State: recommended

Evidence: Deployed runtime code is empty or unverified on target chain.

Recommendation: Deploy and publish verified source code or submit exact deployed bytecode before interaction.

PERMISSION & ROLE GOVERNANCE MAP (PRO)

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Deep analytical map of privileged roles, owner powers, and timelock bounds

- Admin / Owner Interface: NOT ANALYZABLE FROM AVAILABLE EVIDENCE [MISSING]
- Blacklist Capability: NOT ANALYZABLE FROM AVAILABLE EVIDENCE [MISSING]
- Tax / Fee Manipulation: NOT ANALYZABLE FROM AVAILABLE EVIDENCE [MISSING]
- Emergency Pause: NOT ANALYZABLE FROM AVAILABLE EVIDENCE [MISSING]

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- Two-Step Ownership (Ownable2Step): NOT ANALYZABLE FROM AVAILABLE EVIDENCE [MISSING]
- Read-Only Reentrancy Protection: NOT ANALYZABLE FROM AVAILABLE EVIDENCE [MISSING]

Detailed role map traces all execution paths back to verifiable on-chain controllers.

Findings & Vulnerability Analysis:

HIGH

VLM-EVM-NO-CODE: **Unverified On-Chain Code / Missing Bytecode**

Category:

Missing Bytecode Truth | State: open

Evidence:

Deployed runtime code is empty or unverified on target chain.

Recommendation:

Deploy and publish verified source code or submit exact deployed bytecode before interaction.

LIQUIDITY, HOLDER CONCENTRATION & LOCK EVIDENCE (PRO)

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Token distribution curve, whale clusters, and verifiable DEX lock proofs

- Bytecode Footprint: NOT ANALYZABLE FROM AVAILABLE EVIDENCE [MISSING]
- Function Selectors: NOT ANALYZABLE FROM AVAILABLE EVIDENCE [MISSING]
- ERC-20 Conformance: NOT ANALYZABLE FROM AVAILABLE EVIDENCE [MISSING]

Liquidity analysis examines AMM pairing architecture, lock mechanisms, and transaction fee behavior.

ATTACK SURFACE & REENTRANCY FORMAL VECTORS (PRO)

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Simulation of oracle manipulation, flash loans, and cross-function reentrancy

- Oracle Dependency: NOT ANALYZABLE FROM AVAILABLE EVIDENCE [MISSING]
- Flash Loan Attack Vector: NOT ANALYZABLE FROM AVAILABLE EVIDENCE [MISSING]
- Reentrancy Protection: NOT ANALYZABLE FROM AVAILABLE EVIDENCE [MISSING]

STORAGE LAYOUT COLLISION & MULTI-COMPILER BYTECODE DIFF (ADVANCED)

ADVANCED

[LOCKED]

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- Storage slot allocation collision detection across implementation upgrades
- Upgrade proxy layout gap verification and state preservation checks
- Bytecode reproduction against official compiler metadata and build pipelines
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

Protected evidence and findings withheld pending tier upgrade.

FORMAL SMT INVARIANT PROOFS & REGRESSION ANALYSIS (ADVANCED)

ADVANCED

[LOCKED]

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- SMT proofs for solvency, conservation, and reentrancy impossibility invariants
- Bytecode regression analysis and logic change tracking
- Formal SMT Verification (Z3): PROPERTY P VERIFIED UNDER ASSUMPTIONS A FOR MODEL M
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

Protected evidence and findings withheld pending tier upgrade.

HUMAN ANALYST REVIEW & SIGNED VERIFICATION EVIDENCE (ADVANCED)

ADVANCED

[LOCKED]

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

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Summary of Analytical Depth in this section:

- Analyst sign-off requires verified reviewer evidence
- Cryptographically signed attestation digest
- Manual review boundary explicit: no certified safe claims
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).

Protected evidence and findings withheld pending tier upgrade.

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